

Clinical Phenotype Layer v0.2

Expanded phenotype catalog for Honeycomb EHR: standards-backed observables, panel/proxy terms, and local-code gaps.

This catalog contains 134 candidate traits. Interpret coverage status as follows: **Exact standard term** = validated standards-backed concept captured in this pass; **Partial** = broader observable or nearby finding; **Panel/proxy** = standards-backed protocol rather than a perfect one-field code; **Needs local CodeSystem** = likely candidate for Honeycomb or aerospace-local terminology.

FHIR recommendation: prefer **Observation** for measurable phenotype traits, use **Condition/history** for prior events or diagnoses, and create a local CodeSystem for aerospace- or UX-specific traits such as suit fit, ballast requirement, or EVA reach envelope.

Stable phenotype

Trait	Panel	LOINC	SNOMED CT	Status	Recommendation
ABO blood group	Identity & baseline phenotype	883-9 ABO group [Type] in Blood	-	Exact LOINC	Use LOINC for result coding; SNOMED may be used for result answers (e.g., specific blood group findings) rather than as the question code.
Rh factor	Identity & baseline phenotype	10331-7 Rh [Type] in Blood	-	Exact LOINC	Use LOINC for result coding; result answers can use positive/negative answer lists or local binding.
Dominant hand	Identity & baseline phenotype	-	-	Needs local CodeSystem	No high-confidence exact standard code captured in this pass; use local concept now and revisit terminology search later.
Eye color	Visible phenotype	-	-	Needs local CodeSystem	SNOMED editorial guidance explicitly treats color of iris as an observable concept, but no exact production code was validated in this pass.
Hair color	Visible phenotype	-	364547003 Hair observable	Partial SNOMED only	Use local answer set for the result value; the SNOMED code captured here is a broader hair observable, not a precise hair-color question code.
Fitzpatrick skin phototype	Visible phenotype	-	-	Needs local CodeSystem	Create local answer set for Fitzpatrick I-VI; treat as an Observation with ordinal answer values.
Skin undertone	Visible phenotype	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Facial hair / beard pattern	Visible phenotype	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Scar inventory	Visible phenotype	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Tattoo inventory	Visible phenotype	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Freckle distribution	Visible phenotype	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Birthmark inventory	Visible phenotype	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.

Anthropometry

Trait	Panel	LOINC	SNOMED CT	Status	Recommendation
Height	Core anthropometry	8302-2 Body height	50373000 Body height measure	Exact standard term	Use LOINC as Observation.code and SNOMED as alternate semantic mapping if desired.
Weight	Core anthropometry	29463-7 Body weight	27113001 Body weight	Exact standard term	Use LOINC as Observation.code; SNOMED body-weight observable is also available.
Body mass index (BMI)	Core anthropometry	39156-5 Body mass index (BMI) [Ratio]	301331008 Finding of body mass index	Exact LOINC / partial SNOMED	Use LOINC for the measured/calculated Observation; the SNOMED concept captured here is a finding, not a pure observable question code.
Head circumference	Core anthropometry	10173-0 Head circumference	363812007 Head circumference	Exact standard term	Useful for equipment fitting and anthropometric baselining.
Neck circumference	Core anthropometry	56074-8 Circumference Neck	-	Exact LOINC	Good metabolic-risk and suit-fit input.
Waist circumference	Core anthropometry	8280-0 Waist Circumference at umbilicus by Tape measure	-	Exact LOINC	Use local SNOMED mapping later if needed.
Chest circumference	Core anthropometry	8279-2 Chest Circumference at nipple line	-	Exact LOINC	Useful for suit sizing and body-composition tracking.
Arm span	Core anthropometry	56116-7 Width Arm span	-	Exact LOINC	Useful for body proportions and reach-envelope calculations.
Sitting height	Core anthropometry	-	-	Needs local CodeSystem	Useful for cockpit/seat geometry; no exact validated standard code captured in this pass.
Hip circumference	Core anthropometry	-	-	Needs local CodeSystem	Very likely representable with existing standards, but no exact code was validated in this pass.
Shoulder breadth	Core anthropometry	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Hand span	Core anthropometry	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Foot length	Core anthropometry	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Inseam length	Core anthropometry	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Standing reach envelope	Core anthropometry	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Seated reach envelope	Core anthropometry	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Body surface area	Body composition	-	-	Needs local CodeSystem	Use a local Observation concept and record the calculation method (e.g., Mosteller) in Observation.method or note.
Lean body mass	Body composition	-	-	Needs local CodeSystem	May be representable with existing standards, but no exact validated code was captured in this pass.
Fat-free mass	Body composition	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.

Trait	Panel	LOINC	SNOMED CT	Status	Recommendation
Body fat percentage	Body composition	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Visceral fat estimate	Body composition	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Skeletal muscle mass	Body composition	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Frame size	Body composition	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.

Physiology

Trait	Panel	LOINC	SNOMED CT	Status	Recommendation
Heart rate	Baseline vitals	8867-4 Heart rate	-	Exact LOINC	Use for resting HR baseline as well as mission-readiness vitals.
Respiratory rate	Baseline vitals	9279-1 Respiratory rate	-	Exact LOINC	Standard vital sign.
Body temperature	Baseline vitals	8310-5 Body temperature	-	Exact LOINC	Standard vital sign.
Systolic blood pressure	Baseline vitals	8480-6 Systolic blood pressure	-	Exact LOINC	Use with diastolic blood pressure as core panel.
Diastolic blood pressure	Baseline vitals	8462-4 Diastolic blood pressure	-	Exact LOINC	Use with systolic blood pressure as core panel.
Peripheral oxygen saturation	Baseline vitals	59408-5 Oxygen saturation in Arterial blood by Pulse oximetry	-	Exact LOINC	Use pulse-ox specific code where appropriate.
Heart rate variability (R-R interval SD)	Autonomic / wearable	80404-7 R-R interval.standard deviation (Heart rate variability)	-	Exact LOINC	High-value readiness and recovery marker.
Resting heart rate	Autonomic / wearable	8867-4 Heart rate	-	Exact LOINC / local protocol qualifier	Use heart-rate code plus explicit protocol/context indicating resting conditions.
Orthostatic tolerance	Autonomic / wearable	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Heart-rate recovery at 1 minute	Autonomic / wearable	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Sleep duration baseline	Sleep / circadian	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Sleep efficiency baseline	Sleep / circadian	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Circadian midpoint	Sleep / circadian	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Sweat rate	Thermoregulation	-	364538006 Sweating observable	Partial SNOMED only	Use local measurable concept now; SNOMED term captured here is broader than sweat-rate specifically.
Left pupil reaction time	Ocular reflexes	75944-9 Left eye Pupil reaction [Time]	-	Exact LOINC	Useful for neuro checks and mission-readiness monitoring.
Right pupil reaction time	Ocular reflexes	75945-6 Right eye Pupil reaction [Time]	-	Exact LOINC	Useful for neuro checks and mission-readiness monitoring.
Dark adaptation time	Sensory physiology	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Hearing threshold baseline	Sensory physiology	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Vestibular sensitivity baseline	Sensory physiology	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.

Metabolic

Trait	Panel	LOINC	SNOMED CT	Status	Recommendation
Resting metabolic rate (estimated)	Energy expenditure	69429-9 Metabolic rate --resting	-	Exact LOINC	Good fit for energy-expenditure phenotype layer.
Resting metabolic rate (measured)	Energy expenditure	82278-3 Measured RMR	-	Exact LOINC	Use when indirect calorimetry or equivalent measurement is available.
Method used to measure RMR	Energy expenditure	82296-5 Method used to measure RMR	-	Exact LOINC	Store method alongside RMR when you care about measured vs estimated workflows.
Fasting glucose	Glycemic phenotype	1558-6 Fasting glucose [Mass/volume] in Serum or Plasma	-	Exact LOINC	Core metabolic phenotype marker.
Hemoglobin A1c	Glycemic phenotype	4548-4 Hemoglobin A1c/Hemoglobin.total in Blood	-	Exact LOINC	Use U.S. standardized HbA1c code in most U.S. settings.
HOMA-IR	Glycemic phenotype	47214-2 Homeostasis model assessment	-	Exact LOINC	Useful derived phenotype marker when fasting insulin and glucose are available.
Fasting insulin	Glycemic phenotype	-	-	Needs local CodeSystem	Use an exact insulin LOINC in implementation if this becomes core; no high-confidence validation was captured in this pass.
LDL cholesterol	Lipid phenotype	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
HDL cholesterol	Lipid phenotype	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Triglycerides	Lipid phenotype	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Ferritin	Micronutrient phenotype	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Creatinine	Renal / hydration phenotype	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Sodium	Electrolyte phenotype	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Vitamin D	Micronutrient phenotype	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
TSH	Endocrine phenotype	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Sweat sodium sensitivity	Sweat / salt phenotype	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.

Performance

Trait	Panel	LOINC	SNOMED CT	Status	Recommendation
Grip strength - right hand	Strength	83174-3 Grip strength Hand - right Dynamometer	-	Exact LOINC	Best current standards-backed strength marker found in this pass.
Grip strength - left hand	Strength	83173-5 Grip strength Hand - left Dynamometer	-	Exact LOINC	Best current standards-backed strength marker found in this pass.
Grip strength national percentile (dominant hand)	Strength	84419-1 Grip Strength Test - national percentile Hand.dominant [NIH Toolbox]	-	Exact LOINC	Useful when you want normalized rather than raw force.
Grip strength fully adjusted scale score	Strength	84416-7 Grip Strength Test - scale score fully adjusted Hand.dominant [NIH Toolbox]	-	Exact LOINC	Good normalized phenotype signal for longitudinal comparison.
2-minute walk distance	Endurance	83167-7 Walk distance score 2 minutes [NIH Toolbox]	-	Exact LOINC	Usable proxy for endurance when VO2max testing is unavailable.
2-minute walk national percentile	Endurance	84420-9 2-minute Walk Endurance Test - national percentile [NIH Toolbox]	-	Exact LOINC	Normalized endurance marker.
2-minute walk fully adjusted scale score	Endurance	84415-9 2-minute Walk Endurance Test - scale score fully adjusted [NIH Toolbox]	-	Exact LOINC	Normalized endurance marker.
1-mile walk cardiorespiratory fitness protocol	Endurance	62814-9 PhenX - cardiorespiratory fitness - 1 mile walk protocol 150101	-	Panel/proxy standard only	Useful proxy when you want standards-backed protocol support for estimated VO2max rather than direct gas-exchange testing.
Bench press 1RM	Strength	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Squat 1RM	Strength	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Deadlift 1RM	Strength	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Pull-up maximum reps	Strength	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Push-up maximum reps	Strength	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Vertical jump height	Power	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Plank hold duration	Core endurance	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Sprint time 40 m	Speed	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
4-minute mile equivalent	Endurance	-	-	Needs local CodeSystem	Treat as a local derived phenotype; probably too custom for existing terminology.

Trait	Panel	LOINC	SNOMED CT	Status	Recommendation
Agility T-test time	Agility	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Simple reaction time	Dexterity	104888-3 Duration of reaction	-	Proxy LOINC only	This is a generic reaction-duration concept, not a fitness-specific psychomotor battery code.
Tremor amplitude	Dexterity	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Pegboard / fine motor score	Dexterity	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Typing speed	Dexterity	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Tool manipulation score	Task performance	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.

Cognition

Trait	Panel	LOINC	SNOMED CT	Status	Recommendation
MMSE total score	Screening cognition	72106-8 Total score [MMSE]	-	Exact LOINC	Standards-backed global cognitive screen.
MoCA total score	Screening cognition	72172-0 Total score [MoCA]	-	Exact LOINC	Use the total score element rather than the panel when you want a single phenotype field.
Working memory score	Executive / working memory	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Executive function score	Executive / working memory	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Spatial reasoning score	Spatial cognition	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Sustained attention score	Attention	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Processing speed score	Processing speed	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Verbal fluency score	Language	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Stress tolerance score	Operational cognition	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Sleep deprivation resilience score	Operational cognition	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Hazard recognition score	Operational cognition	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Situational awareness score	Operational cognition	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.

Sensory & vestibular

Trait	Panel	LOINC	SNOMED CT	Status	Recommendation
Visual acuity baseline	Visual function	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Contrast sensitivity	Visual function	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Color vision status	Visual function	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Audiometric threshold	Auditory function	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Motion sickness susceptibility	Vestibular function	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Vestibular function score	Vestibular function	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Balance / posturography score	Vestibular function	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Nystagmus burden	Vestibular function	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Migraine susceptibility	Neurologic susceptibility	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Photophobia threshold	Neurologic susceptibility	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.

Aerospace phenotype

Trait	Panel	LOINC	SNOMED CT	Status	Recommendation
Total radiation dose	Radiation	73536-5 Radiation dose total	-	Exact LOINC (from aerospace IG usage)	Directly aligns with the aerospace radiation tracking implementation guide.
Irradiation dose rate	Radiation	77638-4 Irradiation dose rate	-	Exact LOINC (from aerospace IG usage)	Directly aligns with the aerospace radiation tracking implementation guide.
Bone marrow dose	Radiation	-	-	Use aerospace IG / local CodeSystem	The uploaded IG models organ-specific dose components; use the IG/local code system if needed.
Eye lens dose	Radiation	-	-	Use aerospace IG / local CodeSystem	Use local/aerospace IG terminology rather than forcing a generic phenotype code.
Skin dose	Radiation	-	-	Use aerospace IG / local CodeSystem	Use local/aerospace IG terminology rather than forcing a generic phenotype code.
CNS dose	Radiation	-	-	Use aerospace IG / local CodeSystem	Use local/aerospace IG terminology rather than forcing a generic phenotype code.

Trait	Panel	LOINC	SNOMED CT	Status	Recommendation
Suit size class	Suit fit / geometry	-	-	Needs local CodeSystem	Likely best modeled as local operational terminology, not generic SNOMED/LOINC.
Glove size	Suit fit / geometry	-	-	Needs local CodeSystem	Operational fit parameter; local concept preferred.
Helmet size	Suit fit / geometry	-	-	Needs local CodeSystem	Operational fit parameter; local concept preferred.
Neutral buoyancy ballast requirement	Suit fit / geometry	-	-	Use aerospace IG / local CodeSystem	Operational analog-training phenotype; create local concept.
EVA reach envelope	Suit fit / geometry	-	-	Use aerospace IG / local CodeSystem	Mission-design phenotype; local concept preferred.
Dive clearance status	Pressure tolerance	-	-	Use aerospace IG / local CodeSystem	Use the dive medical clearance profile / local terminology stack.
Decompression sickness history	Pressure tolerance	-	-	Use aerospace IG / existing condition coding	Better represented as Condition / history element than as a simple phenotype scalar.
Pressure tolerance score	Pressure tolerance	-	-	Use aerospace IG / local CodeSystem	Derived aerospace phenotype; create local concept.
Neck torque tolerance	Biomechanics	-	-	Needs local CodeSystem	Operational biomechanics concept; local code preferred.
Heat tolerance score	Thermal tolerance	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Cold tolerance score	Thermal tolerance	-	-	Needs local CodeSystem	Create local CodeSystem concept; model as Observation if measurable.
Underwater communication clarity score	Operational performance	-	-	Use aerospace IG / local CodeSystem	Supported conceptually by neutral-buoyancy training metrics, but likely requires local operational terminology.
EVA task performance score	Operational performance	-	-	Use aerospace IG / local CodeSystem	Supported conceptually by neutral-buoyancy training metrics, but likely requires local operational terminology.